



Final Year Project:

Department of Electrical and Electronics Engineering

GSM Signal Denoising and Modulation Classification for Rural Uganda

Improving Connectivity in Underserved Areas Using Machine Learning

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Background

33.2 million Ugandans

exchanged

421.5 million GBs

in 2022

UCC reports that network subscriptions reached **33.2 million** by the end of 2022, and mobile data traffic doubled from 217 million GB in 2020 to 421.5 million GB in 2022 showing an exponential growth in network service demand. [1]

31% of Ugandans

have 4G

62% of Ugandans

Lack mobile internet access

Unfortunately, network expansion is failing to keep pace with this demand; only 31% of the population and 24% of the landmass have 4G coverage and 62% of the population (30 million people) still lacks mobile internet access, and live in low SNR, high-interference environments where the signal is weak (below -90 dBm), the noise becomes louder than the message, making it impossible for standard tools to understand what is being sent[3][4][5][6]

**Most Ugandans still use
basic phones**

Many people in rural Uganda still use basic phones for calls and Mobile Money, but the signal is often very weak and are being blocked or "noised up" by illegal boosters, unlicensed community radios (Bizindaalo), and heavy tropical rain. [7][23][24][25]

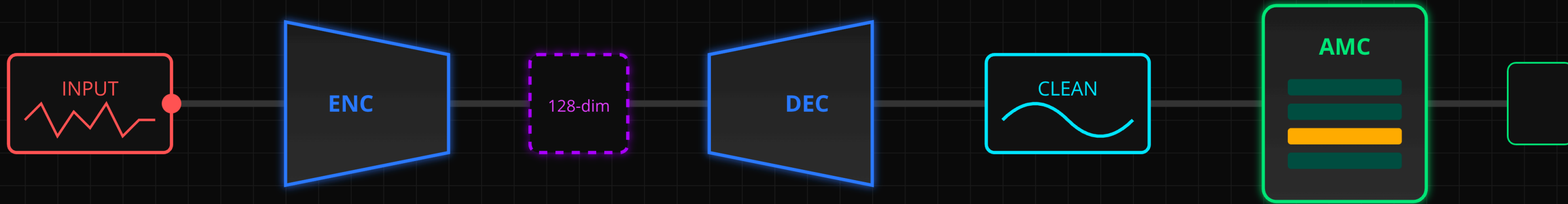
Background continuation

- **Automatic Modulation Classification (AMC)** is crucial for regulators to monitor the spectrum, manage interference, and identify illegal broadcasters that threaten critical services like aviation. [7]
- The project proposes using a **Denoising Autoencoder (DAE)** combined with an **Automatic Modulation Classifier (AMC)** system that takes a noisy signal (e.g., at -90 dBm or even lower) and actively cleans it up removing noise before trying to classify or demodulate it. [5][8][15][26]
- Instead of just trying to read a noisy signal, the proposed system first cleans the signal to make it clear again making sure that even in the most remote sub-counties, the network can still identify and process signals correctly. [8][15][26]



Problem Statement

- UCC's 2024 coverage framework defines 2G/GSM service using a minimum outdoor signal threshold of -90 dBm (RxLEV). However, some rural locations do not consistently meet this threshold. In such weak-signal conditions, current monitoring approaches such as field measurements, drive tests, walk tests, and coverage simulations can confirm weak coverage but can't reliably identify and classify signals because transmissions are buried in interference and noise. [5][27]
- Signal quality is further degraded by illegal boosters, unlicensed community radios (Bizindaalo), and heavy tropical rain. Illegal boosters raise interference, unlicensed transmitters can inject harmful harmonics, and rain fade attenuates the radio path and microwave backhaul. [7][23][24][25]
- UCC therefore finds it difficult to detect illegal broadcasters and diagnose interference when both legitimate and harmful signals are buried in low-SNR, interference-heavy conditions. [7][23][27]
- This leads to dropped calls, failed Mobile Money transactions, and poor service in areas where reliable connectivity is most needed. [3][6][27]
- We therefore lack a robust preprocessing mechanism that can clean weak GSM-family signals before modulation classification and further signal analysis. [8][15][26]



Main Objective

Design a Hybrid DAE-AMC Pipeline

To design, implement, and empirically evaluate a hybrid **Denoising Autoencoder–Automatic Modulation Classification (DAE-AMC) system** that suppresses interference in noisy I/Q streams and maintains accurate modulation decisions across low SNR channel conditions observed in Ugandan networks.

Specific Objectives

01 Design

Define the hybrid DAE-AMC architecture, select the working datasets, and model Uganda-specific low-Signal to Noise Ratio (SNR) interference conditions such as booster oscillation, Bizindaalo harmonics, and rain fade

02 Implementation

Develop and train a one-dimensional convolutional denoising autoencoder, implement the AMC classifier for GSM-family modulations, and integrate both blocks into a single signal-processing pipeline.

03 Analysis

Evaluate the hybrid DAE-AMC pipeline against a standalone AMC baseline using accuracy, F1-score, confusion matrices, and SNR-wise performance trends.

Justification and Significance

Meeting UCC Mandates:

UCC's 2024 coverage framework defines 2G/GSM service using a minimum outdoor signal threshold of -90 dBm (RxLEV), while UCC QoS reporting tracks service thresholds such as blocked and dropped call performance. By targeting weak-signal and interference-heavy conditions around this coverage floor, the proposed DAE-AMC pipeline is aligned with the regulator's need for more reliable monitoring and diagnosis in underserved areas. [5][27]

It Aligns with National Policy:

The proposed ML system supports Uganda's connectivity-expansion agenda by complementing conventional feature-engineered signal analysis with a learnable denoising-and-classification system that is better suited to weak-signal, interference-heavy conditions. [4][15][26]

Spectrum Surveillance

UCC already uses monitoring, measurement, and complaint-handling mechanisms for spectrum oversight. The added value of the proposed system is not basic automation, but low-SNR signal recovery and modulation classification when transmissions are buried in interference and noise. This can strengthen interpretation of weak illegal or harmful transmissions during monitoring and subsequent enforcement analysis, especially for protected services such as aviation and public safety. [7][23][31][32]

Justification and Significance

Critical Services

More reliable weak-signal interpretation can support voice communication, Mobile Money, and other essential services that depend on stable rural mobile links. [5][6]

Rural Mobile Connectivity

The direct focus of this project is reliable rural mobile connectivity rather than broadband throughput alone. By improving signal interpretation near the coverage floor, the proposed system supports more stable GSM/2G service conditions in underserved areas where weak coverage and interference coexist. [3][5][6][27]

Scope

1. Offline Experimentation

Project conducted entirely on a computer using pre-existing data, ensuring a controlled environment.

2. Datasets

DeepSig RadioML 2018.01A benchmark data provides synthetic, well-characterized SNR labels, while the RF Signal Data collection on Kaggle provides real SDR captures that introduce hardware impairment realism. [11][12]

3. Noise Modelling

We inject Uganda-specific noise profiles such as Booster oscillation noise as High-power wideband impulses simulating illegal repeater feedback, Bizindaalo harmonics as Narrowband tones at 2nd/3rd harmonics of FM frequencies (200 MHz, 300 MHz sidebands leaking into 900 MHz), Rain fade attenuation as Time-varying SNR reduction modeling tropical storm conditions (100+ mm/h rainfall rates)

4. Model Development

Focus on a Conv1D Denoising Autoencoder (DAE) and a supervised AMC classifier.

Specific Modulations

Training models to recognize GSM-family modulations like GMSK, GFSK, and QPSK

Platform & Tools

Python, PyTorch, Jupyter Notebook, and supporting visualization or cloud tools where needed.

Timeline

A 6-month dedicated period for research, model development, and comprehensive testing.

Literature Review

Chapter Two



Classical Methods

Early Automatic Modulation Classification (AMC) relied mainly on statistical analysis, likelihood-based decision rules, and manually engineered signal descriptors such as higher-order moments and cyclostationary properties. These approaches are analytically elegant and can perform well in controlled settings, but their performance degrades sharply when channel conditions are unknown or when the received signal is weak and interference-heavy

Key Limitation:

- Classical AMC methods depend on stable channel assumptions and carefully engineered signal descriptors.
- In congested and interference-affected deployments such as those documented by UCC, weak-signal and noise-heavy conditions make those descriptors less reliable, motivating the transition toward data-driven learning approaches. [7][23][27]

Machine learning as the new standard

Modern research has shifted to deep learning models such as CNNs, Transformers, and Mixture-of-Experts frameworks. These models can learn signal characteristics directly from raw signal data and have shown significant accuracy improvements, especially in low-SNR conditions down to 0 dB. [15][16][26]

GSM Noise Mitigation in African Networks

While DAE-based signal enhancement has been extensively studied academically, its application to African GSM networks' specific interference environment remains unexplored. [8][26]

Regulatory Interventions

UCC's zero-tolerance posture has led to enforcement operations that confiscate illegal boosters and shut down unlicensed Bizindaalo stations. These interventions are important for spectrum control, but they mainly address interference after it has already been detected. [7][23]

Current Technical Practice

In practice, operators and regulators rely on coverage measurements, drive tests, walk tests, field troubleshooting, and related monitoring procedures to identify weak coverage and interference sources. These methods are useful for confirming degraded service conditions, but they do not directly recover weak GSM signals once those signals are buried in interference and noise. [27][31][32]

The Research Gap

Although machine-learning methods for denoising and modulation classification are well studied, there is still limited work applying them to interference mitigation and weak-signal GSM monitoring in African networks, especially for Uganda-specific conditions such as booster oscillation, Bizindaalo harmonics, and tropical rain fade. [7][8][23][24][25][26]

We address that gap by training a DAE on synthetic noise profiles matching documented interference vectors, enabling GMSK classification at SNR levels where conventional approaches fail. [8][15][26]

Machine Learning in AMC

Deep learning has reshaped AMC by learning discriminative signal characteristics directly from raw I/Q samples. Recent contributions demonstrate tangible low-SNR gains:

AbdElaziz et al. (2023)

1

CNN with parallel asymmetric kernels achieved 96.5% accuracy at 0 dB and 86.1% at -2 dB across nine modulations. [15]

2

Zhang et al. (2023)

MoE-AMC mixture-of-experts framework yielded ~71.8% averaged accuracy across -20...18 dB. [16]

Meta-Transformer (2024)

3

Leverages transformer encoders and few-shot learning to adapt rapidly to unseen modulations while maintaining superior accuracy. [30]

4

Rehman et al. (2025)

DL AMC converts I/Q streams into eye diagrams, overcoming the 10–48% accuracy ceiling near -10 dB SNR.

Denoising Autoencoders for Signal Enhancement

Denoising frontends have emerged as an effective countermeasure when raw I/Q features are overwhelmed by interference.

75%

77.5%

70%

Classification accuracy across –12...8 dB .

Dual residual DAE with channel attention improved AMC accuracy by up to 75% across –12...8 dB SNR. [8]

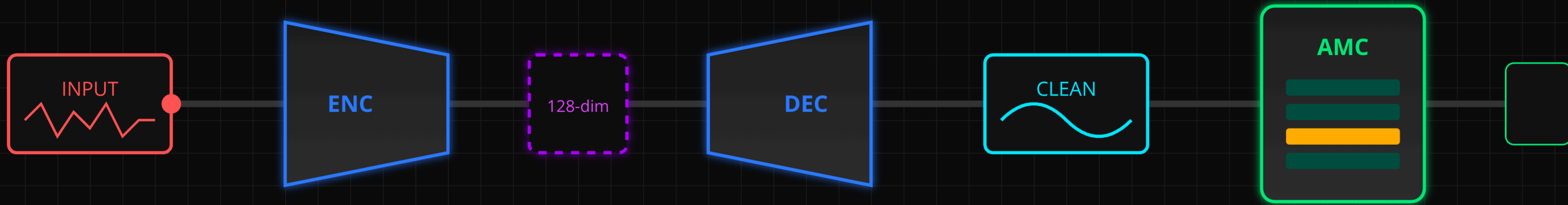
accuracy at –10 dB

Faysal et al.'s multimodal denoising masked autoencoder sustained 77.5% accuracy at -10 dB, roughly 22% higher than the non-denoised case. [28]

relative accuracy gains on low-SNR samples.

An and Lee's thresholded autoencoder with an SNR predictor delivered about 70% relative accuracy gains on low-SNR samples. [29]

These findings justify the DAE-AMC architecture explored in this project and provide design cues for gating strategies that conserve energy on SDR deployments. [8][26][28][29]



CHAPTER THREE – METHODOLOGY

A comprehensive framework for developing robust Automatic Modulation Classification systems tailored to challenging RF environments through deep learning and denoising techniques.

Introduction to Methods

We adopt an experimental research design anchored in reproducible data processing and quantitative benchmarking which is why our methodology is organized into three linked parts: **design**, **implementation**, and **analysis**. [11][12][15][16][26]

01 DESIGN

This stage defines the full architecture and experiment setup before training begins.

-  - Select the datasets to be used, primarily RF Signal Data and RadioML 2018.01A. [11][12]
-  - Harmonize all signals into a unified tensor format with 1024-sample windows and normalized amplitude.
-  - Perform exploratory data analysis (EDA) to inspect class balance, SNR coverage, waveform quality, and impairment patterns before training. [11][12]
-  - Define the GSM-focused modulation classes: GMSK, GFSK, and QPSK.
-  - Model Uganda-specific interference using booster oscillation, Bizindaalo harmonics, and rain-fade attenuation. [7][23][24][25]
-  - Specify the system architecture showing how the denoising block feeds the modulation classifier.

Expected outcome: A complete project design package comprising the architecture diagram, dataset specification, low-SNR scenario definitions, and the experimental setup for the DAE-AMC pipeline.

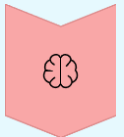
Methodology

02 IMPLEMENTATION

This stage converts the design into working models and an integrated signal-processing pipeline.



- Train a Conv1D encoder-decoder DAE on noisy and clean I/Q pairs using MSE-based reconstruction loss. [8][26][28]



- Train a supervised AMC classifier on GSM-family modulations using both raw and denoised inputs. [15][16]



- Integrate the DAE and AMC blocks into a hybrid pipeline and compare frozen and fine-tuned configurations. [8][26][28][29]

- Build a lightweight prototype interface for visualizing noisy signals, denoised outputs, and classification predictions.

Expected outcome: A functional hybrid DAE-AMC prototype capable of denoising noisy I/Q samples and producing modulation predictions under controlled low-SNR conditions.

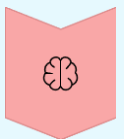
Methodology

03 ANALYSIS AND EVALUATION

This stage measures whether the proposed pipeline improves modulation recognition under the targeted interference conditions.



- Benchmark the hybrid DAE-AMC system against a standalone AMC baseline. [15][16]



- Evaluate denoising quality and classification quality across held-out SNR ranges. [8][26][28][29]



- Use accuracy curves, macro F1-score, confusion matrices, and SNR-wise comparisons to quantify improvement. [15][16][26]
- Interpret the results in terms of rural Uganda use cases such as weak-signal monitoring, interference diagnosis, and regulator demonstrations.

Expected outcome: A clear evidence set showing where the hybrid pipeline improves over baseline AMC, including result plots, comparative metrics, and conclusions about suitability for rural Uganda spectrum-monitoring scenarios.

Project Budget Overview

The successful execution of this project requires a structured allocation of resources. The total estimated budget is **1,750,000 UGX**, covering critical areas from software to research.

Item	Cost (UGX)
Software Licenses	300,000
Data Curation	100,000
Nvidia GPUs	500,000
Cloud Databases	100,000
Internet and Research	200,000
Printing and Documentation	50,000
Miscellaneous	500,000
Total Estimated Cost	1,750,000

This budget ensures that all necessary technical infrastructure, data processing capabilities, and operational requirements are adequately funded to support the project's objectives.

Project Timeline

Our project will be executed in five distinct phases, ensuring a structured and efficient progression towards our objectives over a 27-week period.

Phase	MONTH 1				MONTH 2				MONTH 3				MONTH 4				MONTH 5				MONTH 6			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Problem Framing and Literature Review																								
Data Acquisition, EDA, and Preprocessing																								
Model Design and Architecture																								
Implementation and testing																								
Evaluation and Analysis																								
Documentation and Submission																								

This timeline ensures a methodical approach to each stage, culminating in a robust and well-documented solution for intelligent signal processing.

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Thank you for Listening